

Name \_\_\_\_\_

**SHORT ANSWER.** Write the word or phrase that best completes each statement or answers the question.

Approximate the number using a calculator. Round your answer to three decimal places (2pts each).

1)  $4^{3.9}$  1) \_\_\_\_\_

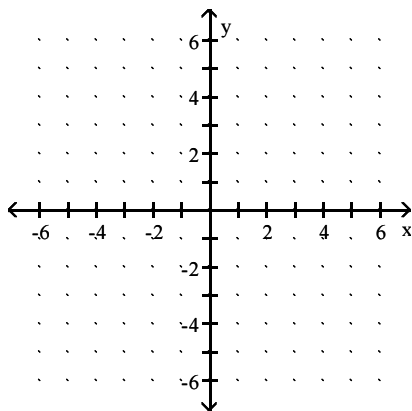
2)  $3.6\pi$  2) \_\_\_\_\_

**Solve the problem.**

- 3) The function  $f(t) = 500(0.5)^{t/80}$  models the amount in pounds of a particular radioactive material stored in a concrete vault, where  $x$  is the number of years since the material was put into the vault. Find the amount of radioactive material in the vault after 60 years. Round to the nearest whole number. 3) \_\_\_\_\_

**Graph the function by making a table of coordinates.**

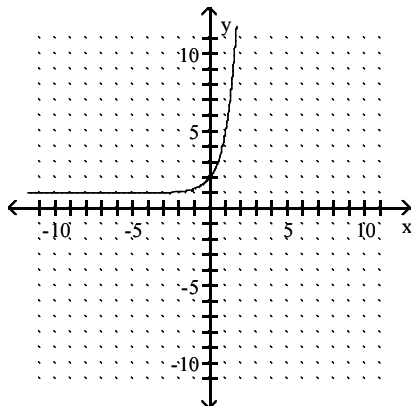
4)  $f(x) = 4^x$  4) \_\_\_\_\_



**MULTIPLE CHOICE.** Choose the one alternative that best completes the statement or answers the question.

The graph of an exponential function is given. Select the function for the graph from the functions listed (6 pts).

5) 5) \_\_\_\_\_



A)  $f(x) = 4^x$

B)  $f(x) = 4^x - 1$

C)  $f(x) = 4^x + 1$

D)  $f(x) = 4^x + 1$

**SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.**

Use the compound interest formulas  $A = P\left(1 + \frac{r}{n}\right)^{nt}$  and  $A = Pe^{rt}$  to solve.

- 6) Find the accumulated value of an investment of \$7000 at 5% compounded monthly for 7 years. 6) \_\_\_\_\_

**Write the equation in its equivalent exponential form.**

- 7)  $\log_2 x = 3$  7) \_\_\_\_\_

- 8)  $\log_b 64 = 3$  8) \_\_\_\_\_

**Write the equation in its equivalent logarithmic form.**

- 9)  $4^3 = 64$  9) \_\_\_\_\_

- 10)  $2^{-2} = \frac{1}{4}$  10) \_\_\_\_\_

- 11)  $\sqrt[3]{64} = 4$  11) \_\_\_\_\_

Evaluate the expression without using a calculator.

12)  $\log_3 1$

12) \_\_\_\_\_

13)  $\log_{11} 11$

13) \_\_\_\_\_

14)  $\log_3 3^{17}$

14) \_\_\_\_\_

15)  $e^{\ln 136}$

15) \_\_\_\_\_

Find the domain of the logarithmic function.

16)  $f(x) = \log_6 (x + 3)$

16) \_\_\_\_\_

17)  $f(x) = \ln (2 - x)$

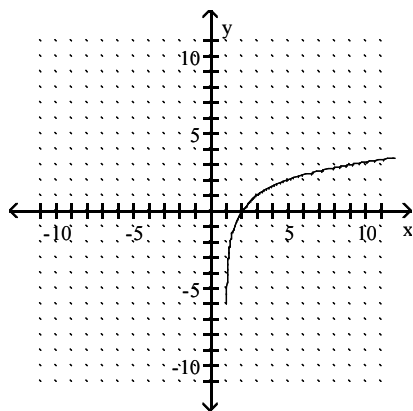
17) \_\_\_\_\_

**MULTIPLE CHOICE.** Choose the one alternative that best completes the statement or answers the question.

The graph of a logarithmic function is given. Select the function for the graph from the options.

18)

18) \_\_\_\_\_



A)  $f(x) = \log_2 (x - 1)$

C)  $f(x) = \log_2 (x + 1)$

B)  $f(x) = \log_2 x - 1$

D)  $f(x) = \log_2 x$

**SHORT ANSWER.** Write the word or phrase that best completes each statement or answers the question.

Use properties of logarithms to expand the logarithmic expression as much as possible. Where possible, evaluate logarithmic expressions without using a calculator.

19)  $\log_2 (7 \cdot 3)$  19) \_\_\_\_\_

20)  $\log_6 \left( \frac{2}{5} \right)$  20) \_\_\_\_\_

21)  $\log_5 13^{-2}$  21) \_\_\_\_\_

22)  $\ln(5 \cdot e^3)$  22) \_\_\_\_\_

Use properties of logarithms to condense the logarithmic expression. Write the expression as a single logarithm whose coefficient is 1. Where possible, evaluate logarithmic expressions.

23)  $\log_5 10 - \log_5 2$  23) \_\_\_\_\_

24)  $3 \log_x 2 + \log_x 2$  24) \_\_\_\_\_

25)  $\ln x + 6 \ln y$  25) \_\_\_\_\_